

Optimization for Data Science, Autumn 2026 - Final Project

For your final project, you will work in groups of three to six students on a research question broadly related to optimization for data science. Your question could be theoretical, methodological, or related to an application. The project consists of three deliverables: (i) a short document describing your question and team, (ii) a final report, and (iii) an in-person presentation.

The central goal is to develop your ability to identify and formulate research questions worth pursuing. Read relevant papers, identify a gap in the literature, and work toward answering a precise question over the quarter. A complete solution or a publishable paper is not required. A compelling question and meaningful partial results can make a strong project. In what follows, we elaborate on the deliverables, the timeline, and interesting/relevant papers for several topics.

1 Deliverables

All documents listed below have to use an 11pt font and 1in margins.

1.1 Question and team document

Due: October 26, 2026.

Submit to: Gradescope.

What to submit: Submit a short PDF (2 pages max) with one submission per team and all team members identified. Include your team members' names, a working project title, and a precise statement of your proposed question. Briefly explain why the question matters, how it connects to the papers you have read, and what the successful execution of the project would look like. Cite the relevant papers. This document establishes your proposed direction. Note that the question may be refined as you learn more. You will get feedback from Mateo.

1.2 Final report

Due: December 12, 2026.

Submit to: Gradescope.

What to submit: A report (5 pages max, not counting references) with an appendix if necessary (20 pages max). Your report should explain the question, its significance, what was already known, and what your team accomplished. The appendix can contain additional details that did not fit in the main body. Organize the report so that you address the following points.

1. **Question and motivation.** State the question clearly and explain why it matters.
2. **Related work and formulation.** Describe the closest relevant work and define the problem, notation, and assumptions.
3. **Approach and results.** Present your methods, analysis, algorithms, proofs, counterexamples, or experiments. Distinguish established results from your own contribution and provide evidence for your claims.
4. **Discussion and limitations.** Explain what the results establish, which assumptions matter, and what remains unresolved. Carefully supported partial or negative results can be valuable.

5. **References and supporting material.** Cite sources and provide enough detail to assess your reasoning and reproduce computational results where applicable.

1.2.1 How will we grade the report?

There are three criteria for grading this report.

1. **Writing quality.** Your report has to be easy to read and very, VERY clear. Be careful when using LLMs because they tend to use a lot of (made-up) jargon and write nonsensical paragraphs that are nearly impossible for humans to parse. If your report reads like that, do not expect a high grade.
2. **Content.** We expect your question to be relevant and your approach to answering it to be somewhat novel and interesting. If we learn something new while reading your report, even if it comes only from the basic question and a partial approach, you will get full marks here.
3. **One-shottable.** We will try to answer your question using ChatGPT 6.0 Astra with one prompt. If it generates a report that is on par with or better than your report (in terms of the previous two criteria), do not expect a high grade.

1.3 Presentation

Schedule: December 7, 2026 (tentative).

Format: In person. Each talk will be between 15 and 30 minutes (per group). Additional details about the exact location and time will be disclosed closer to the end of the quarter.

1.3.1 How will we assess the presentation?

We expect you to explain the question, why it is interesting, your approach, and your main results and limitations. Choose material that makes the central contribution clear within the allotted time. Every team member must understand and be able to explain every aspect of the project; your understanding will be assessed during the presentation and discussion.

2 How to find good research questions?

1. **Start from a few papers.** Read closely enough to understand the problem, the main contribution, and the evidence supporting it. The topics and papers below offer starting points.
2. **Examine assumptions and limitations.** Identify which conditions a result needs and what the analysis or experiments leave unexplained. Separate the authors' claims from your own interpretation.
3. **Formulate a precise question.** State what you want to understand and what evidence would count as an answer. Define the setting and keep the scope feasible within the quarter.
4. **Check the surrounding literature.** Follow references and later papers that cite the work. Verify whether the question has already been answered before claiming a gap.

5. **Make an initial attempt.** Work through a small example, proof argument, or baseline experiment. Use what you learn to refine the question and assess whether the project is feasible.

We cannot emphasize this enough: Meet with the instructor periodically for feedback on your question, scope, and progress.

3 What makes a good research question?

This is a hard question to answer, and it is one that most LLMs cannot actually answer. But let's give it a try! I find it easier to answer this question by looking at the two extremes. Excellent questions (and their answers) usually start new subfields (like the ones that you will see in the list below). Usually this is because: (i) they develop methods/ideas that allow us to tackle problems that seemed impossible before, or (ii) they illuminate our understanding of a phenomenon, which unlocks new research directions. At the opposite end of the spectrum, incremental papers give us ϵ -improvements in what we can solve and in our understanding. Usually, incremental questions slightly modify a method, or the assumptions of a result, but do not truly change the spirit of what we were already doing. Most, though not all, questions of the type *What if I combine idea A with idea B?* are in this category.

With all that said, finding good questions and having 'good taste' are skills that take years to refine (and most of us keep on working on them). The goal here is not for you to identify and solve an award-winning question, but rather to start developing that muscle.

4 AI Policy

AI use is encouraged in this project. A central goal is to learn to use these tools effectively. You may use them to explore ideas, clarify papers, assist with implementation, and improve exposition. You remain responsible for the correctness and clarity of the submitted work. Check mathematical claims, code, experimental results, and citations. Your report must present your team's original ideas or results and accurately represent your contribution. Give credit for ideas, text, code, and other materials drawn from external sources, including AI tools that contributed to the work. Every team member is responsible for understanding and being able to explain every aspect of the project. AI assistance does not replace that responsibility; your understanding will be assessed during the final presentation.

5 Deadlines and extensions

Each student has a one-day, no-questions-asked extensions for homework assignments or the final report. Extensions apply to separate submissions and cannot be combined into a two-day extension. Team members' extensions do not compound: the final project can be submitted at most **one day late** using these extensions. For illness, emergencies, or other circumstances affecting your work, contact the instructor as soon as reasonably possible.

6 Topics and relevant papers

The following collection contains **222 distinct papers across 57 topics**. You may choose a related topic beyond this collection. Use the papers to start understanding the literature and develop your own research question. For many of these topics, you will need to read/understand a wide range of papers (likely way beyond whatever we include here) to find good questions. This is a daunting task, but there is only one way to learn. Good luck!

As a word of warning, we include established and more recent subfields. For mature fields, researchers have been working for long enough that most relevant questions are well understood, and the ones that are not are widely regarded as crucial questions, e.g., the Riemann hypothesis. Fields that are still in their infancy tend to have more fertile ground, so many natural questions are still up for grabs.

I. Scaling, post-training, generation, and representation learning

1. Scaling laws, compute-optimal training, and uncertainty in extrapolation

- [Jared Kaplan et al. \(2020\)](#), [Scaling Laws for Neural Language Models](#) — *foundation*. Fits empirical relationships between language-model loss, parameters, data and training compute, turning resource allocation and finite stopping time into quantitative modeling questions. The reported allocation depends on the study’s training protocol.
- [Jordan Hoffmann et al. \(2022\)](#), [Training Compute-Optimal Large Language Models](#) — *method*. Revises compute-optimal model/data allocation through a large training sweep and the Chinchilla experiment; under the studied protocol, optimal parameters and training tokens grow approximately together as compute increases.
- [Tomer Poria et al. \(2024\)](#), [Resolving Discrepancies in Compute-Optimal Scaling of Language Models](#) — *method*. Shows how output-layer compute, warmup and scale-dependent optimizer tuning explain conflicting compute-optimal predictions in reproduced language-model experiments. Also investigates learning-rate, batch-size and AdamW beta2 scaling.
- [Yasaman Bahri et al. \(2024\)](#), [Explaining Neural Scaling Laws](#) — *theory*. Distinguishes variance-limited and resolution-limited scaling mechanisms, connecting model/data scaling to infinite-width limits, kernel spectra and data geometry in controlled settings.
- [Tamay Besiroglu et al. \(2024\)](#), [Chinchilla Scaling: A replication attempt](#) — *replication*. Re-examines Chinchilla’s parametric fitting procedure using measurements reconstructed from plots, identifying fitting and uncertainty issues; its rederived allocation agrees with the other two estimation procedures. Useful for distinguishing a fitted curve from a reliable extrapolation.

2. Inference and test-time compute: search, sampling and budget allocation

- [Xuezhi Wang et al. \(2023\)](#), [Self-Consistency Improves Chain of Thought Reasoning in Language Models](#) — *foundation*. Samples multiple reasoning paths and aggregates answers instead of using greedy decoding. Provides a simple inference-budget baseline; correlated wrong answers limit what agreement alone can certify.

- Charlie Snell et al. (2025), [Scaling LLM Test-Time Compute Optimally Can be More Effective than Scaling Parameters for Reasoning](#) — *method*. Studies verifier-guided search and adaptive response revision under inference budgets. Demonstrates why the best allocation depends on problem difficulty and why inference compute is not universally interchangeable with pretraining.
- Niklas Muennighoff et al. (2025), [s1: Simple test-time scaling](#) — *method*. Combines a small curated reasoning-trace dataset with budget forcing to control generation length. Distinguishes inference intervention, supervised training and actual computation cost.

3. Preference optimization and reasoning post-training: RLHF, DPO and GRPO

- Nisan Stiennon et al. (2020), [Learning to summarize with human feedback](#) — *foundation*. Trains a reward model from human summary comparisons and optimizes a language-model policy. Provides a concrete example of mismatch between reference-likelihood training, automatic metrics and the desired evaluation objective.
- Rafael Rafailov et al. (2023), [Direct Preference Optimization: Your Language Model is Secretly a Reward Model](#) — *method*. Reparameterizes KL-regularized reward optimization into a preference-classification loss under the stated model. Makes reference policies, preference data and regularization strength explicit optimization choices.
- Zichen Liu et al. (2025), [Understanding R1-Zero-Like Training: A Critical Perspective](#) — *method*. Examines pretrained-model effects and GRPO normalization biases, proposing Dr. GRPO to improve token efficiency. Separates objective design from observed reasoning behavior in verifiable-reward training.
- Zhihong Shao et al. (2024), [DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models](#) — *method*. Introduces Group Relative Policy Optimization (GRPO), estimating a baseline from grouped responses to reduce critic-model memory; combines mathematical pretraining with reinforcement learning and studies mathematical evaluation tasks.

4. Modern generative objectives: diffusion, flow matching and consistency models

- Jonathan Ho et al. (2020), [Denoising Diffusion Probabilistic Models](#) — *foundation*. Connects diffusion training to denoising score matching through a weighted variational objective. Exposes timestep weighting, prediction targets and sampling discretization as distinct design choices.
- Yaron Lipman et al. (2023), [Flow Matching for Generative Modeling](#) — *method*. Trains continuous normalizing flows through conditional vector-field regression without simulating trajectories during training. Connects probability-path design and transport-inspired interpolation to optimization and numerical sampling cost.
- Yang Song et al. (2023), [Consistency Models](#) — *method*. Learns mappings consistent along probability-flow trajectories, supporting one- or few-step generation. Separates standalone consistency training from distillation and makes teacher error and discretization explicit modeling and numerical variables.

5. Self-supervised representation learning: contrastive, covariance and predictive objectives

- Ting Chen et al. (2020), [A Simple Framework for Contrastive Learning of Visual Representations](#) — *foundation*. Introduces SimCLR and studies augmentations, projection heads and batch sizes in contrastive learning. A clear baseline for analyzing stochastic objectives and representation quality under limited computation.
- Adrien Bardes et al. (2022), [VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning](#) — *method*. Uses explicit variance, invariance and covariance penalties to learn representations while avoiding constant embeddings. Offers a tractable objective for studying regularization, finite-batch statistics and collapse.
- Mahmoud Assran et al. (2023), [Self-Supervised Learning From Images With a Joint-Embedding Predictive Architecture](#) — *method*. Predicts target-region representations from image context rather than reconstructing pixels. Makes masking, target networks and prediction-space objectives explicit variables in scalable self-supervised optimization.

II. Contemporary training methods, data, and architectures

6. Modern optimizer design: matrix preconditioning, orthogonalized updates, and sign momentum

- Nikhil Vyas et al. (2025), [SOAP: Improving and Stabilizing Shampoo using Adam for Language Modeling](#) — *method*. Runs Adam in Shampoo’s evolving preconditioner eigenbasis, connecting matrix preconditioning and adaptive moments; language-model experiments study the efficiency cost of infrequent eigendecomposition.
- Jingyuan Liu et al. (2025), [Muon is Scalable for LLM Training](#) — *method*. Studies scaling matrix-orthogonalized Muon updates to large language models through weight decay and update-scale choices; its efficiency comparisons are empirical and training-setting dependent.
- Xiangning Chen et al. (2023), [Symbolic Discovery of Optimization Algorithms](#) — *method*. Searches symbolic update programs and introduces Lion, a sign-based momentum optimizer with one state buffer; reported gains depend on retuning learning rate and weight decay.

7. Feature-learning parameterizations, μ P, and hyperparameter transfer

- Greg Yang and Edward J. Hu (2021), [Tensor Programs IV: Feature Learning in Infinite-Width Neural Networks](#) — *theory*. Classifies width-dependent parameterizations and establishes feature-learning infinite-width limits, providing the theoretical foundation for maximal-update parameterization rather than only kernel-limit training.
- Greg Yang et al. (2021), [Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer](#) — *method*. Uses μ P to transfer hyperparameters tuned on smaller-width networks to larger networks; width stability does not automatically imply transfer across depth, architecture, or training horizon.

- Soufiane Hayou (2026), [A Proof of Learning Rate Transfer Under \$\mu P\$ — theory](#). Proves convergence of the loss-minimizing learning rate to a positive, width-independent limit for deep linear networks trained by gradient descent with squared loss and fixed input/output layers. Gives an explicit one-step limit; at later fixed training steps, transfer assumes a unique minimizer of the limiting loss.

8. Schedule-free learning, learning-rate adaptation, and training-horizon dependence

- Aaron Defazio et al. (2024), [The Road Less Scheduled — method](#). Unifies scheduling and iterate averaging to develop schedule-free optimization without requiring a preset stopping horizon; a learning-rate hyperparameter and method-specific averaging choices remain.
- Aaron Defazio and Konstantin Mishchenko (2023), [Learning-Rate-Free Learning by D-Adaptation — method](#). Adapts an unknown distance-to-solution scale, proving an asymptotically optimal convex Lipschitz guarantee and proposing practical SGD and Adam variants evaluated empirically in deep learning.
- Maor Ivgi et al. (2023), [DoG is SGD’s Best Friend: A Parameter-Free Dynamic Step Size Schedule — method](#). Introduces Distance over Gradients (DoG), adapting SGD step sizes from distance to initialization and accumulated gradient norms without a tuned learning-rate parameter. A tamed variant has high-probability convergence and iterate-movement guarantees in stochastic convex optimization; practical and per-layer variants are evaluated on vision and language transfer-learning tasks.
- Shane Bergsma et al. (2025), [Straight to Zero: Why Linearly Decaying the Learning Rate to Zero Works Best for LLMs — method](#). Compares decay schedules for LLM pretraining and explains linear decay-to-zero through an AdamW averaging view; its favorable results require tuned peak rates and the studied budgets.

9. Gradient-noise scales, critical batch size, and optimization hyperparameter scaling laws

- Sam McCandlish et al. (2018), [An Empirical Model of Large-Batch Training — foundation](#). Proposes the gradient noise scale as an empirical predictor of useful batch size and models compute-versus-time efficiency; the proxy requires assumptions and is not a universal identity.
- William Merrill et al. (2025), [Critical Batch Size Revisited: A Simple Empirical Approach to Large-Batch Language Model Training — method](#). Directly measures critical batch size rather than inferring it from gradient noise; OLMo experiments show early growth and later saturation, motivating batch-size warmup.
- Shane Bergsma et al. (2025), [Power Lines: Scaling laws for weight decay and batch size in LLM pre-training — method](#). Fits empirical LLM scaling laws linking weight decay, batch size, learning rate, model size, and token budget; optimal and critical batch-size claims are specific to the tested training setup.

10. Optimizing pretraining data mixtures and domain sampling

- Sang Michael Xie et al. (2023), [DoReMi: Optimizing Data Mixtures Speeds Up Language Model Pretraining](#) — *method*. Learns domain sampling weights with group distributionally robust optimization on a small proxy, then transfers the mixture to larger-model pretraining; connects competing domain losses to data allocation.
- Simin Fan et al. (2024), [DOGE: Domain Reweighting with Generalization Estimation](#) — *method*. Uses a bilevel proxy-training algorithm to learn domain sampling weights from estimated generalization effects, then transfers them to larger models; provides a useful comparison with minimax data reweighting.

11. Data-constrained training, data reuse and synthetic-data feedback

- Niklas Muennighoff et al. (2023), [Scaling Data-Constrained Language Models](#) — *method*. Studies language-model scaling when unique data are limited, modeling diminishing returns from repeated data to guide allocation between model size and training passes under a compute budget.
- Ilya Shumailov et al. (2024), [AI models collapse when trained on recursively generated data](#) — *counterexample*. Shows loss of distributional information under studied recursive synthetic-data training protocols, including tail disappearance; motivates analyzing statistical feedback separately from optimizing each generation’s training loss.
- Matthias Gerstgrasser et al. (2024), [Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data](#) — *counterexample*. Contrasts replacing data with accumulating original and generated samples; reports avoided degradation in studied experiments and bounded error in a linear model, challenging an unconditional interpretation of model collapse.

12. Mixture-of-experts routing, capacity constraints and load balancing

- William Fedus et al. (2022), [Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity](#) — *foundation*. Simplifies sparse expert routing and introduces training choices addressing communication, load balance and numerical instability; illustrates that total parameter count and per-token active computation differ.
- Lean Wang et al. (2024), [Auxiliary-Loss-Free Load Balancing Strategy for Mixture-of-Experts](#) — *method*. Updates expert-specific routing biases from recent loads to balance utilization without adding an auxiliary-loss gradient to task training; compares the resulting load/performance tradeoff with penalty-based balancing.

13. Memory and low-rank optimization for large models

- Noam Shazeer and Mitchell Stern (2018), [Adafactor: Adaptive Learning Rates with Sub-linear Memory Cost](#) — *method*. Factors second-moment estimates and combines update clipping with adaptive scaling to reduce optimizer-state memory for Transformer training.
- Samyam Rajbhandari et al. (2020), [ZeRO: Memory Optimizations Toward Training Trillion Parameter Models](#) — *method*. Partitions redundant optimizer states, gradients and

parameters across workers, making memory–communication tradeoffs part of large-model optimization.

- Edward J. Hu et al. (2022), [LoRA: Low-Rank Adaptation of Large Language Models](#) — *method*. Restricts adaptation to trainable low-rank factors while freezing pretrained weights, offering a tractable study of parameterization, rank and optimization.
- Jiawei Zhao et al. (2024), [GaLore: Memory-Efficient LLM Training by Gradient Low-Rank Projection](#) — *method*. Projects gradients into changing low-rank subspaces to shrink optimizer states while permitting full-parameter updates.
- Tim Dettmers et al. (2023), [QLoRA: Efficient Finetuning of Quantized LLMs](#) — *method*. Back-propagates through frozen quantized base weights into low-rank adapters, combining low-bit storage, double quantization and paged optimizers to reduce finetuning memory. Separates quantization error, adaptation capacity and optimizer memory.
- Shih-Yang Liu et al. (2024), [DoRA: Weight-Decomposed Low-Rank Adaptation](#) — *method*. Separates weight magnitude and direction during adaptation, using low-rank updates for the directional component; studies differences from full finetuning and reports improvements over LoRA in the evaluated tasks.

III. Training dynamics, generalization, and scientific machine learning

14. Large learning rates, catapult dynamics, and self-stabilization at the edge of stability

- Jeremy Cohen et al. (2021), [Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability](#) — *method*. Documents full-batch neural training near the Hessian stability threshold, where short-term loss increases coexist with longer-term progress; an empirical challenge to monotone-descent intuition.
- Aitor Lewkowycz et al. (2020), [The large learning rate phase of deep learning: the catapult mechanism](#) — *theory*. Analyzes solvable neural training dynamics where large learning rates cause transient loss growth and curvature reduction, supported by neural experiments; behavior depends on the model and rate regime.
- Alex Damian et al. (2023), [Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability](#) — *theory*. Explains self-stabilization through coupled iterate and sharpness dynamics, relating gradient descent to a sharpness-constrained trajectory under stated local assumptions; it is not a universal convergence theorem.

15. Heavy-tailed gradient noise, clipping, and relaxed smoothness

- Jingzhao Zhang et al. (2020), [Why Gradient Clipping Accelerates Training: A Theoretical Justification for Adaptivity](#) — *theory*. Introduces gradient-dependent smoothness and analyzes clipped/normalized updates, showing that this alternative curvature model can favor adaptivity over fixed-step gradient descent.
- Jingzhao Zhang et al. (2020), [Why are Adaptive Methods Good for Attention Models?](#) — *theory*. Connects heavy-tailed stochastic-gradient noise with adaptive/clipped training

through finite-moment convergence analysis and attention-model experiments; proposes coordinate-wise clipping.

- Savelii Chezhegov et al. (2025), [Clipping Improves Adam-Norm and AdaGrad-Norm when the Noise Is Heavy-Tailed](#) — *theory*. Gives poor high-probability convergence examples for norm-based adaptive methods under heavy-tailed noise and establishes improved guarantees with clipping, including delayed-step variants.

16. Benign overfitting, double descent and grokking as distinct training phenomena

- Peter L. Bartlett et al. (2020), [Benign overfitting in linear regression](#) — *theory*. Characterizes when minimum-norm interpolation of noisy linear data predicts accurately through covariance effective ranks; overparameterization alone does not ensure benign overfitting.
- Preetum Nakkiran et al. (2020), [Deep Double Descent: Where Bigger Models and More Data Hurt](#) — *counterexample*. Documents model-size and training-time double descent and regimes where additional data worsen test performance; proposes effective model complexity as a unifying empirical perspective, not a general theorem.
- Neel Nanda et al. (2023), [Progress measures for grokking via mechanistic interpretability](#) — *method*. Reverse engineers Fourier-based modular addition in small transformers and tracks memorization, circuit formation and cleanup; offers measurable training progress underlying delayed generalization in this studied task.

17. Sharpness, normalization and training geometry

- Nitish Shirish Keskar et al. (2017), [On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima](#) — *foundation*. Provides influential empirical evidence connecting large batches, sharpness and generalization; read with reparameterization critiques rather than as a universal causal theorem.
- Laurent Dinh et al. (2017), [Sharp Minima Can Generalize For Deep Nets](#) — *counterexample*. Constructs function-preserving reparameterizations that change sharpness, showing why parameter-space flatness alone cannot explain generalization.
- Shibani Santurkar et al. (2018), [How Does Batch Normalization Help Optimization?](#) — *theory*. Investigates optimization smoothness as an explanation for batch normalization’s effectiveness and challenges the internal-covariate-shift account.
- Pierre Foret et al. (2021), [Sharpness-Aware Minimization for Efficiently Improving Generalization](#) — *method*. Optimizes a neighborhood worst-case loss with a practical two-gradient approximation, connecting training geometry to robustness and generalization experiments.
- Sergey Ioffe and Christian Szegedy (2015), [Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift](#) — *method*. Introduces normalization within neural-network training. Later work re-examines its internal-covariate-shift explanation through optimization geometry.

- Jeremy Cohen et al. (2021), Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability — *method*. Documents full-batch neural training near the Hessian stability threshold, where short-term loss increases coexist with longer-term progress; an empirical challenge to monotone-descent intuition.

18. Neural tangent kernels, mean-field limits and feature learning

- Arthur Jacot et al. (2018), Neural Tangent Kernel: Convergence and Generalization in Neural Networks — *foundation*. Describes neural-network gradient-flow dynamics through a kernel that becomes deterministic and fixed in an infinite-width regime.
- Song Mei et al. (2018), A mean field view of the landscape of two-layer neural networks — *theory*. Derives a distributional PDE description of two-layer neural-network SGD under a mean-field scaling, offering a feature-learning counterpart to fixed-kernel analysis.
- Simon S. Du et al. (2019), Gradient Descent Provably Optimizes Over-parameterized Neural Networks — *theory*. Proves global linear convergence for sufficiently wide two-layer ReLU networks with squared loss under input and initialization assumptions.
- Lénaïc Chizat et al. (2019), On Lazy Training in Differentiable Programming — *counterexample*. Shows that parameter scaling can induce nearly linearized training and gives experiments cautioning against treating the lazy regime as a universal explanation of deep-learning performance.

19. Scientific machine learning: PINN conditioning and composite residual losses

- Maziar Raissi et al. (2019), Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations — *foundation*. Develops neural training objectives combining observed data with differential-equation residuals for forward and inverse problems, supplying a concrete setting for composite losses and automatic differentiation.
- Sifan Wang et al. (2021), Understanding and Mitigating Gradient Flow Pathologies in Physics-Informed Neural Networks — *method*. Diagnoses stiffness and imbalance among back-propagated gradients in composite PINN losses, then uses gradient statistics for adaptive loss balancing and studies a more resilient architecture.
- Pratik Rathore et al. (2024), Challenges in Training PINNs: A Loss Landscape Perspective — *theory*. Relates differential-operator ill-conditioning to the PINN training landscape, compares Adam and L-BFGS combinations, and develops a Nyström-preconditioned Newton-CG method.

20. Machine unlearning: deletion updates, retraining and optimization state

- Chuan Guo et al. (2020), Certified Data Removal from Machine Learning Models — *foundation*. Defines certified removal relative to models trained without deleted data and develops a mechanism for linear classifiers. Provides a precise alternative to interpreting poor forget-set accuracy as successful deletion.

- Seth Neel et al. (2021), [Descent-to-Delete: Gradient-Based Methods for Machine Unlearning — theory](#). Studies sequential deletion for convex models and distinguishes indistinguishability of internal optimization state from observable outputs. Useful for reasoning about momentum, warm starts and repeated deletion requests.
- Meghdad Kurmanji et al. (2023), [Towards Unbounded Machine Unlearning — method](#). Introduces SCRUB and evaluates neural unlearning with application-dependent forgetting and utility metrics. Encourages explicit retraining comparisons and cautions against treating a single forgetting metric as a complete guarantee.

21. Bilevel optimization, hypergradients and differentiable solvers

- Dougal Maclaurin et al. (2015), [Gradient-based Hyperparameter Optimization through Reversible Learning — foundation](#). Differentiates through training trajectories to optimize hyperparameters, using reversible momentum dynamics to reduce storage.
- Brandon Amos and J. Zico Kolter (2017), [OptNet: Differentiable Optimization as a Layer in Neural Networks — method](#). Places quadratic programs inside neural networks and obtains gradients through sensitivity analysis and implicit differentiation of optimality conditions.
- Luca Franceschi et al. (2018), [Bilevel Programming for Hyperparameter Optimization and Meta-Learning — theory](#). Unifies hyperparameter tuning and meta-learning as bilevel problems and analyzes approximation through finite inner optimization dynamics.
- Jonathan Lorraine et al. (2020), [Optimizing Millions of Hyperparameters by Implicit Differentiation — method](#). Uses implicit differentiation and inverse-Hessian approximations to compute hypergradients without storing an entire optimization trajectory.

22. Inverse problems, algorithm unrolling, and learned optimizers

- Karol Gregor and Yann LeCun (2010), [Learning Fast Approximations of Sparse Coding — foundation](#). Learns fixed-depth approximations to sparse-coding solvers. Training cost, inference accuracy, and changes in the measurement operator or signal distribution expose the limits of amortized optimization.
- Marcin Andrychowicz et al. (2016), [Learning to learn by gradient descent by gradient descent — method](#). Trains recurrent networks to produce optimization updates on a task distribution. Generalization to new objective scales, dimensions, horizons, and problem families should be evaluated separately from training-task performance.
- Luke Metz et al. (2019), [Understanding and correcting pathologies in the training of learned optimizers — method](#). Diagnoses biased or unstable gradients from truncated optimizer unrolling and proposes alternative estimators, addressing meta-training reliability and practical performance.
- Vishal Monga et al. (2021), [Algorithm Unrolling: Interpretable, Efficient Deep Learning for Signal and Image Processing — survey](#). Reviews unrolling methods, applications, and theoretical limitations, including data consistency, robustness, and convergence outside the training distribution.

23. Benign landscapes, PL inequalities, and overparameterization

- Kenji Kawaguchi (2016), [Deep Learning without Poor Local Minima — theory](#). Characterizes deep linear-network squared-loss landscapes and distinguishes favorable local minima from degenerate saddles; nonlinear conclusions require additional assumptions.
- Hamed Karimi et al. (2016), [Linear Convergence of Gradient and Proximal-Gradient Methods Under the Polyak-Łojasiewicz Condition — theory](#). Uses the PL inequality and a proximal generalization to obtain linear convergence without strong convexity, including selected nonconvex problems.
- Simon S. Du et al. (2019), [Gradient Descent Finds Global Minima of Deep Neural Networks — theory](#). Proves convergence to zero training loss for sufficiently overparameterized deep networks under specified initialization and data conditions, emphasizing Gram-matrix stability and residual architectures.
- Zeyuan Allen-Zhu et al. (2019), [A Convergence Theory for Deep Learning via Over-Parameterization — theory](#). Establishes polynomial-time training guarantees for sufficiently wide deep networks, including ReLU architectures, by controlling the neighborhood around random initialization.
- Samet Oymak and Mahdi Soltanolkotabi (2019), [Overparameterized Nonlinear Learning: Gradient Descent Takes the Shortest Path? — theory](#). Under local Jacobian conditioning and regularity around initialization, obtains geometric least-squares convergence and path-length bounds; the shortest-path claim concerns near-minimality rather than exact optimality.

24. Implicit bias and optimization-dependent solution selection

- Daniel Soudry et al. (2018), [The Implicit Bias of Gradient Descent on Separable Data — theory](#). Shows directional convergence of gradient descent on separable linear logistic regression to a maximum-margin predictor, despite diverging parameter norms and slow margin convergence.
- Suriya Gunasekar et al. (2018), [Implicit Bias of Gradient Descent on Linear Convolutional Networks — theory](#). Under directional-convergence assumptions, links full-width linear convolutional networks to Fourier-domain bridge penalties; for deeper networks the guarantee concerns first-order stationarity, not global penalty minimization.
- Suriya Gunasekar et al. (2018), [Characterizing Implicit Bias in Terms of Optimization Geometry — theory](#). Studies implicit solution selection for mirror, natural, and steepest descent in underdetermined linear models and separable classification, connecting optimizer geometry to regularization.
- Ziwei Ji and Matus Telgarsky (2019), [The implicit bias of gradient descent on nonseparable data — theory](#). Decomposes logistic-regression dynamics into a maximum-margin direction on a maximal separable subset and a bounded offset describing remaining observations; the full dataset need not be separable.

- Kaifeng Lyu and Jian Li (2020), [Gradient Descent Maximizes the Margin of Homogeneous Neural Networks](#) — *theory*. Connects homogeneous-network training to normalized-margin dynamics and KKT points of a margin problem; KKT stationarity need not imply global margin maximization.

25. Algorithmic stability, early stopping, and optimization–statistical tradeoffs

- Olivier Bousquet and André Elisseeff (2002), [Stability and Generalization](#) — *foundation*. Formalizes algorithmic stability and derives generalization bounds, providing a route from sensitivity to training-data changes toward predictive guarantees.
- Moritz Hardt et al. (2016), [Train faster, generalize better: Stability of stochastic gradient descent](#) — *theory*. Bounds SGD generalization through stability under Lipschitz and smoothness assumptions, relating step sizes and iteration counts to algorithm-dependent statistical performance.
- Garvesh Raskutti et al. (2014), [Early Stopping and Non-parametric Regression: An Optimal Data-dependent Stopping Rule](#) — *theory*. Derives a data-dependent early-stopping rule for kernel least squares with minimax-optimal statistical rates and a connection to kernel ridge regression.
- Léon Bottou and Olivier Bousquet (2008), [The Tradeoffs of Large Scale Learning](#) — *foundation*. Separates approximation, estimation, and optimization error and explains why the statistically useful target accuracy can change which optimization method is preferable.

IV. Distributed and federated optimization

26. Federated learning and heterogeneous local objectives

- H. Brendan McMahan et al. (2017), [Communication-Efficient Learning of Deep Networks from Decentralized Data](#) — *foundation*. Introduces federated averaging: clients perform local SGD and periodically average models, making communication and heterogeneous data central optimization concerns.
- Tian Li et al. (2020), [Federated Optimization in Heterogeneous Networks](#) — *method*. Adds proximal regularization to local objectives and accommodates variable local work, linking statistical heterogeneity with device constraints.
- Sai Praneeth Karimireddy et al. (2020), [SCAFFOLD: Stochastic Controlled Averaging for Federated Learning](#) — *method*. Uses control variates to correct client drift and supplies theory explaining why heterogeneous local updates can slow FedAvg.
- Sashank J. Reddi et al. (2021), [Adaptive Federated Optimization](#) — *method*. Adapts server-side Adagrad, Adam and Yogi to federated training and analyzes their interaction with heterogeneous client data.

27. Distributed and decentralized optimization

- Angelia Nedić and Asuman Ozdaglar (2009), [Distributed Subgradient Methods for Multi-Agent Optimization](#) — *foundation*. Combines local subgradient steps and neighbor averaging; establishes convergence and accuracy–iteration tradeoffs under network assumptions.

- John C. Duchi et al. (2012), [Dual Averaging for Distributed Optimization: Convergence Analysis and Network Scaling](#) — *theory*. Separates optimization error from communication effects and relates convergence to network spectral gaps, providing a useful bridge from convex methods to graph-dependent performance.
- Wei Shi et al. (2015), [EXTRA: An Exact First-Order Algorithm for Decentralized Consensus Optimization](#) — *method*. Corrects the steady-state bias of decentralized gradient descent and permits constant-step convergence to the consensus optimum under stated assumptions.
- Xiangru Lian et al. (2017), [Can Decentralized Algorithms Outperform Centralized Algorithms? A Case Study for Decentralized Parallel Stochastic Gradient Descent](#) — *method*. Studies decentralized SGD through both convergence analysis and systems experiments, showing why communication bottlenecks can change the best architecture.
- Stephen Boyd et al. (2011), [Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers](#) — *survey*. Connects ADMM, operator splitting and distributed convex statistical learning, with implementations and examples; a useful bridge between centralized optimization and networked computation.

28. Gradient compression and communication efficiency

- Dan Alistarh et al. (2017), [QSGD: Communication-Efficient SGD via Gradient Quantization and Encoding](#) — *method*. Quantizes gradients with convergence guarantees, making the tradeoff between transmitted bits and estimator variance explicit.
- Jeremy Bernstein et al. (2018), [signSGD: Compressed Optimisation for Non-Convex Problems](#) — *method*. Transmits gradient signs and studies the geometry and noise assumptions under which sign-based updates and majority voting work.
- Sai Praneeth Karimireddy et al. (2019), [Error Feedback Fixes SignSGD and other Gradient Compression Schemes](#) — *counterexample*. Constructs failure examples for biased compression and shows how retaining compression residuals can restore SGD-like convergence under the paper’s assumptions.
- Thijs Vogels et al. (2019), [PowerSGD: Practical Low-Rank Gradient Compression for Distributed Optimization](#) — *method*. Combines low-rank gradient approximation, error feedback and all-reduce-compatible communication, emphasizing wall-clock performance rather than compression ratio alone.

29. Asynchronous and delayed stochastic optimization

- Feng Niu et al. (2011), [Hogwild!: A Lock-Free Approach to Parallelizing Stochastic Gradient Descent](#) — *foundation*. Analyzes shared-memory SGD without locking, using sparse updates to control interference and motivate the tradeoff between synchronization overhead and optimization error.
- Alekh Agarwal and John C. Duchi (2011), [Distributed Delayed Stochastic Optimization](#) — *theory*. Bounds the effect of stale stochastic gradients and shows when smoothness makes bounded delays asymptotically negligible.

- Xiangru Lian et al. (2015), [Asynchronous Parallel Stochastic Gradient for Nonconvex Optimization](#) — *theory*. Extends asynchronous SGD analysis to nonconvex objectives, relating permissible worker counts and delays to convergence and potential speedup.
- Rémi Leblond et al. (2017), [ASAGA: Asynchronous Parallel SAGA](#) — *theory*. Combines asynchronous updates with variance reduction and revisits a subtle sampling/read-order issue in asynchronous convergence proofs.

V. Decisions, robustness, privacy, and probabilistic learning

30. Decision-focused learning and performative prediction

- Priya L. Donti et al. (2017), [Task-based End-to-end Model Learning in Stochastic Optimization](#) — *method*. Trains predictive models against downstream optimization objectives through an end-to-end stochastic programming formulation.
- Adam N. Elmachtoub and Paul Grigas (2022), [Smart “Predict, then Optimize”](#) — *method*. Introduces an optimization-aware prediction loss and a tractable surrogate that account for downstream decision costs.
- Juan Perdomo et al. (2020), [Performative Prediction](#) — *foundation*. Formalizes prediction-induced distribution changes and distinguishes stable retraining fixed points from performative optima.
- John P. Miller et al. (2021), [Outside the Echo Chamber: Optimizing the Performative Risk](#) — *theory*. Shows why stable retraining can be suboptimal and exploits structural assumptions to optimize risk under endogenous distribution shift.

31. Distributionally robust and adversarial optimization

- Aharon Ben-Tal et al. (2013), [Robust Solutions of Optimization Problems Affected by Uncertain Probabilities](#) — *foundation*. Derives tractable robust counterparts for probability uncertainty described by divergence regions and connects those regions to statistical confidence sets.
- Hongseok Namkoong and John C. Duchi (2016), [Stochastic Gradient Methods for Distributionally Robust Optimization with f-divergences](#) — *method*. Develops stochastic algorithms for divergence-based robust empirical risk objectives and calibrated performance guarantees.
- Aman Sinha et al. (2018), [Certifying Some Distributional Robustness with Principled Adversarial Training](#) — *method*. Uses a penalized Wasserstein robustness formulation to couple parameter updates with adversarial data perturbations under smoothness assumptions.
- John C. Duchi and Hongseok Namkoong (2021), [Learning Models with Uniform Performance via Distributionally Robust Optimization](#) — *theory*. Studies statistical and optimization guarantees for robust objectives aimed at distribution shifts and heterogeneous subpopulation performance.

32. Differential privacy and constrained stochastic optimization

- Kamalika Chaudhuri et al. (2011), [Differentially Private Empirical Risk Minimization — foundation](#). Develops privacy-preserving regularized empirical risk minimization through perturbation, showing how privacy constraints alter the optimization problem.
- Raef Bassily et al. (2014), [Private Empirical Risk Minimization: Efficient Algorithms and Tight Error Bounds — theory](#). Establishes algorithms and matching error bounds for private convex empirical risk minimization under Lipschitzness and bounded-domain assumptions.
- Martín Abadi et al. (2016), [Deep Learning with Differential Privacy — method](#). Introduces scalable private deep-learning techniques with gradient clipping, noise and refined privacy accounting.
- Raef Bassily et al. (2019), [Private Stochastic Convex Optimization with Optimal Rates — theory](#). Obtains optimal excess population-loss rates up to logarithmic factors, connecting stability, privacy and stochastic convex optimization.

33. Fairness constraints and optimization of nondifferentiable metrics

- Muhammad Bilal Zafar et al. (2017), [Fairness Constraints: Mechanisms for Fair Classification — method](#). Develops constrained optimization mechanisms for balancing classification accuracy with a tractable measure of decision-boundary fairness.
- Alekh Agarwal et al. (2018), [A Reductions Approach to Fair Classification — method](#). Reduces constrained fair classification to cost-sensitive classification and a game, enabling optimization with multiple fairness constraints.
- Michael Kearns et al. (2018), [Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness — theory](#). Exposes failures of fairness guarantees on a few predefined groups and develops auditing and learning procedures for richer subgroup families.
- Andrew Cotter et al. (2019), [Optimization with Non-Differentiable Constraints with Applications to Fairness, Recall, Churn, and Other Goals — method](#). Uses proxy constraints and game-based optimization to address nondifferentiable rate constraints, with attention to feasibility and generalization.

34. Robust statistics, contamination and optimization certificates

- Ilias Diakonikolas et al. (2016), [Robust Estimators in High Dimensions without the Computational Intractability — foundation](#). Develops efficient high-dimensional robust estimation with dimension-independent error for specified distribution families; motivates filtering and optimization-based estimators.
- Kevin A. Lai et al. (2016), [Agnostic Estimation of Mean and Covariance — method](#). Estimates moments under malicious contamination with polynomial-time algorithms, connecting robust statistics, covariance geometry and spectral methods.
- Yu Cheng et al. (2019), [High-Dimensional Robust Mean Estimation in Nearly-Linear Time — method](#). Uses primal-dual semidefinite formulations to obtain nearly-linear-time robust mean estimation under specified tail or covariance assumptions.

35. Multiobjective optimization and multitask learning

- Jean-Antoine Désidéri (2012), [Multiple-gradient descent algorithm \(MGDA\) for multiobjective optimization](#) — *foundation*. Constructs common descent directions for multiple smooth objectives and analyzes convergence to Pareto-stationary points.
- Ozan Sener and Vladlen Koltun (2018), [Multi-Task Learning as Multi-Objective Optimization](#) — *method*. Applies multiobjective optimization to shared neural representations, addressing trade-offs between competing task losses.
- Tianhe Yu et al. (2020), [Gradient Surgery for Multi-Task Learning](#) — *method*. Projects conflicting task gradients to reduce interference in multitask training; useful for examining geometry, scaling and order dependence.

36. Policy gradients and reinforcement-learning optimization

- Richard S. Sutton et al. (1999), [Policy Gradient Methods for Reinforcement Learning with Function Approximation](#) — *foundation*. Derives a policy-gradient representation suitable for estimation with compatible function approximation, linking stochastic optimization and reinforcement learning.
- Sham M. Kakade (2001), [A Natural Policy Gradient](#) — *method*. Uses policy-space geometry to precondition gradients and relates natural-gradient updates to policy improvement.
- John Schulman et al. (2015), [Trust Region Policy Optimization](#) — *method*. Develops a trust-region policy improvement scheme and a practical approximate implementation; useful for studying where theory and implemented updates diverge.
- Alekh Agarwal et al. (2021), [On the Theory of Policy Gradient Methods: Optimality, Approximation, and Distribution Shift](#) — *theory*. Characterizes global convergence, approximation error and sampling requirements for policy gradients under specified policy classes and distribution-shift conditions.

37. Variational inference and optimization of probabilistic models

- Matthew D. Hoffman et al. (2013), [Stochastic Variational Inference](#) — *method*. Uses stochastic optimization for scalable approximate Bayesian inference, with natural-gradient updates in structured latent-variable models.
- Diederik P. Kingma and Max Welling (2014), [Auto-Encoding Variational Bayes](#) — *method*. Introduces reparameterized stochastic variational learning and amortized inference for continuous latent-variable generative models.
- Danilo Jimenez Rezende et al. (2014), [Stochastic Backpropagation and Approximate Inference in Deep Generative Models](#) — *method*. Derives stochastic backpropagation for scalable joint optimization of generative models and recognition networks.
- David M. Blei et al. (2017), [Variational Inference: A Review for Statisticians](#) — *survey*. Reviews approximate Bayesian inference as optimization, emphasizing statistical properties, approximation choices and stochastic computational methods.

38. Langevin sampling, Bayesian learning and score-based generation

- [Max Welling and Yee Whye Teh \(2011\), Bayesian Learning via Stochastic Gradient Langevin Dynamics — *foundation*](#). Combines minibatch stochastic gradients with injected noise to approximate Bayesian sampling, connecting optimization updates with posterior exploration.
- [Arnak S. Dalalyan \(2017\), Theoretical guarantees for approximate sampling from smooth and log-concave densities — *theory*](#). Provides nonasymptotic sampling guarantees for Langevin methods under smooth log-concavity, separating sampling error from optimization error.
- [Maxim Raginsky et al. \(2017\), Non-convex learning via Stochastic Gradient Langevin Dynamics: a nonasymptotic analysis — *theory*](#). Analyzes finite-time empirical and population optimization guarantees for noisy nonconvex learning under regularity and dissipativity conditions.
- [Yang Song and Stefano Ermon \(2019\), Generative Modeling by Estimating Gradients of the Data Distribution — *method*](#). Learns scores across noise levels and uses annealed Langevin dynamics for generation; motivates studying score error, geometry and discretization.

39. Online learning, regret and parameter-free optimization

- [Martin Zinkevich \(2003\), Online Convex Programming and Generalized Infinitesimal Gradient Ascent — *foundation*](#). Establishes projected online gradient descent and regret guarantees against a fixed comparator; a starting point for changing objectives and sequential decision-making.
- [Elad Hazan et al. \(2007\), Logarithmic regret algorithms for online convex optimization — *theory*](#). Shows how curvature, including exp-concavity, permits logarithmic regret and motivates online Newton steps.
- [Shai Shalev-Shwartz \(2012\), Online Learning and Online Convex Optimization — *survey*](#). Organizes online learning through regret, regularization and convexity; useful for distinguishing an online guarantee from a statistical generalization guarantee.
- [Francesco Orabona and Dávid Pál \(2016\), Coin Betting and Parameter-Free Online Learning — *method*](#). Uses coin betting to construct online algorithms that adapt to unknown comparator norms without an oracle learning-rate choice.

40. Bandit feedback and derivative-free optimization

- [Abraham D. Flaxman et al. \(2005\), Online Convex Optimization in the Bandit Setting: Gradient Descent Without a Gradient — *foundation*](#). Builds gradient estimates from function evaluations and randomized smoothing, connecting bandit feedback to first-order optimization.
- [Ohad Shamir \(2013\), On the Complexity of Bandit and Derivative-Free Stochastic Convex Optimization — *theory*](#). Provides lower bounds and achievable rates that expose dimension-dependent limitations of derivative-free stochastic optimization.

- John C. Duchi et al. (2015), [Optimal Rates for Zero-Order Convex Optimization: The Power of Two Function Evaluations](#) — *theory*. Contrasts paired function evaluations with weaker feedback models and quantifies the dimension cost of gradient estimation.
- Yurii Nesterov and Vladimir Spokoiny (2017), [Random gradient-free minimization of convex functions](#) — *theory*. Analyzes randomized gradient-free methods using function values and smoothing; separates smoothness assumptions from oracle access.

41. Minimax optimization, variational inequalities and GAN training

- Arkadi Nemirovski (2004), [Prox-Method with Rate of Convergence \$O\(1/t\)\$ for Variational Inequalities with Lipschitz Continuous Monotone Operators and Smooth Convex-Concave Saddle Point Problems](#) — *foundation*. Introduces Mirror-Prox for monotone variational inequalities and convex-concave games; a foundation for understanding extragradient corrections.
- Ian J. Goodfellow et al. (2014), [Generative Adversarial Nets](#) — *foundation*. Frames generative modeling as a two-player minimax game, exposing the difference between optimizing a scalar loss and finding an equilibrium.
- Constantinos Daskalakis et al. (2018), [Training GANs with Optimism](#) — *method*. Studies optimistic updates that counter cycling, proving last-iterate convergence for a class of bilinear games and testing GAN applications.
- Lars Mescheder et al. (2018), [Which Training Methods for GANs do actually Converge?](#) — *counterexample*. Uses simple GAN counterexamples to expose nonconvergence and identifies local convergence conditions for instance noise and zero-centered gradient penalties.

VI. Structured models, computational geometry, and discrete optimization

42. Optimal transport, Sinkhorn scaling, and Wasserstein computation

- Jean-David Benamou and Yann Brenier (2000), [A computational fluid mechanics solution to the Monge-Kantorovich mass transfer problem](#) — *foundation*. Reformulates quadratic optimal transport dynamically through density and momentum, solved by an augmented-Lagrangian method. Discretization and mass-conservation constraints connect PDE modeling with convex optimization.
- Marco Cuturi (2013), [Sinkhorn Distances: Lightspeed Computation of Optimal Transport](#) — *method*. Uses entropy regularization and matrix scaling to make transport computation practical. Regularization introduces a tradeoff among speed, numerical conditioning, approximation, and the meaning of the resulting comparison.
- Jason Altschuler et al. (2017), [Near-linear time approximation algorithms for optimal transport via Sinkhorn iteration](#) — *theory*. Analyzes Sinkhorn-based approximate transport and introduces Greenkhorn coordinate updates. Near-linearity refers to problem-input size with accuracy-dependent factors; feasibility correction and regularization choices matter.

- Gabriel Peyré and Marco Cuturi (2019), [Computational Optimal Transport: With Applications to Data Science](#) — *survey*. Surveys computational transport, entropic methods, barycenters, and data-science applications, organizing computational formulations and their connections.

43. Riemannian and manifold optimization

- Alan Edelman et al. (1998), [The Geometry of Algorithms with Orthogonality Constraints](#) — *foundation*. Derives optimization algorithms on Stiefel and Grassmann manifolds. The geometric perspective clarifies feasible updates, tangent spaces, and symmetries in eigen-vector estimation and subspace learning.
- Silvére Bonnabel (2013), [Stochastic Gradient Descent on Riemannian Manifolds](#) — *method*. Extends stochastic gradient descent to manifold-valued variables and studies convergence to critical points. Useful for covariance estimation, subspace learning, and examining geometric assumptions behind stochastic algorithms.
- Bart Vandereycken (2013), [Low-Rank Matrix Completion by Riemannian Optimization](#) — *method*. Optimizes over the fixed-rank matrix manifold using Riemannian conjugate gradients, retractions, and vector transport.
- Nicolas Boumal et al. (2019), [Global rates of convergence for nonconvex optimization on manifolds](#) — *theory*. Establishes iteration complexity for approximate first- and second-order stationarity using Riemannian gradient and trust-region methods. Guarantees concern stationarity, not global optimality; issue year 2019, online publication 2018.

44. Randomized numerical linear algebra, sketching, data selection, and coresets

- Nathan Halko et al. (2011), [Finding Structure with Randomness: Probabilistic Algorithms for Constructing Approximate Matrix Decompositions](#) — *survey*. Surveys and analyzes randomized range-finding and low-rank decompositions. Spectral decay, oversampling, passes over data, and orthogonalization stability determine accuracy and practical efficiency.
- Kenneth L. Clarkson and David P. Woodruff (2017), [Low-Rank Approximation and Regression in Input Sparsity Time](#) — *theory*. Develops sparse embeddings for regression and low-rank approximation with input-sparsity runtime. Projects should include sketch construction, failure probability, and downstream solution accuracy in their cost accounting.
- Olivier Bachem et al. (2017), [Practical Coreset Constructions for Machine Learning](#) — *survey*. Explains sensitivity sampling and coreset guarantees for clustering and other learning problems. A coreset approximates a specified family of objectives; it does not automatically preserve every downstream task.
- Baharan Mirzasoleiman et al. (2020), [Coresets for Data-efficient Training of Machine Learning Models](#) — *method*. Selects weighted training subsets to approximate full gradients using submodular optimization. Updating subsets, selection overhead, and changes in model gradients affect whether data reduction yields end-to-end savings.

45. Low-rank matrix completion, factorization, and robust PCA

- Benjamin Recht et al. (2010), [Guaranteed Minimum-Rank Solutions of Linear Matrix Equations via Nuclear Norm Minimization](#) — *theory*. Extends sparse-recovery geometry to rank minimization. Nuclear-norm recovery guarantees depend on measurement conditions; this makes a useful comparison point for scalable nonconvex factorization methods.
- Emmanuel J. Candès and Benjamin Recht (2009), [Exact Matrix Completion via Convex Optimization](#) — *foundation*. Establishes exact recovery of low-rank matrices from randomly observed entries under incoherence conditions. Nonuniform missingness and coherent singular vectors are natural stress tests of those assumptions.
- Emmanuel J. Candès et al. (2011), [Robust Principal Component Analysis?](#) — *method*. Uses a convex low-rank-plus-sparse decomposition to recover components under incoherence and corruption assumptions. Identifiability can fail when low-rank structure and sparse contamination overlap.
- Rong Ge et al. (2016), [Matrix Completion has No Spurious Local Minimum](#) — *theory*. Proves benign geometry for a specified positive-semidefinite matrix-completion formulation under sampling and structural assumptions. The result is not a universal guarantee for arbitrary recommender-system data.

46. Phase retrieval and nonlinear inverse-problem landscapes

- Emmanuel J. Candès et al. (2013), [PhaseLift: Exact and Stable Signal Recovery from Magnitude Measurements via Convex Programming](#) — *foundation*. Lifts phaseless measurements into semidefinite optimization with exact and stable recovery guarantees for random measurements. Matrix lifting trades favorable convex geometry for substantial memory and computation.
- Emmanuel J. Candès et al. (2015), [Phase Retrieval via Wirtinger Flow: Theory and Algorithms](#) — *method*. Combines spectral initialization with inexpensive nonconvex updates for phaseless recovery. Projects can separate the roles of initialization, measurement distribution, noise, and local refinement.
- Praneeth Netrapalli et al. (2013), [Phase Retrieval using Alternating Minimization](#) — *method*. Analyzes alternating phase and signal updates under Gaussian measurements. Inspect the resampling used in theoretical analysis before comparing guarantees with implementations that reuse one fixed dataset.
- John C. Duchi and Feng Ruan (2019), [Solving \(most\) of a set of quadratic equalities: composite optimization for robust phase retrieval](#) — *method*. Studies robust phase retrieval through nonsmooth composite objectives and prox-linear updates, including faulty measurements. Connects sharpness and weak convexity with recovery; issue year 2019, online publication 2018.

47. Tensor decomposition and latent-variable estimation

- Tamara G. Kolda and Brett W. Bader (2009), [Tensor Decompositions and Applications](#) — *survey*. Introduces CP, Tucker, and related tensor models with applications and algorithms.

A practical entry point for understanding how higher-order factorization differs from matrix SVD.

- Christopher J. Hillar and Lek-Heng Lim (2013), [Most Tensor Problems Are NP-Hard — counterexample](#). Shows hardness of many general tensor analogues of tractable matrix problems. Any proposed efficient global method must specify which structural assumptions avoid these worst-case barriers.
- Animashree Anandkumar et al. (2014), [Tensor Decompositions for Learning Latent Variable Models — method](#). Uses structured moment tensors and a robust tensor power method for latent-variable estimation. Whitening, orthogonality, and moment-estimation error provide concrete assumptions to examine.
- Rong Ge et al. (2015), [Escaping From Saddle Points — Online Stochastic Gradient for Tensor Decomposition — theory](#). Connects stochastic saddle escape to orthogonal tensor decomposition through strict-saddle geometry. Offers a tractable structured case to contrast with general tensor hardness.

48. Submodular and discrete optimization for data science

- George L. Nemhauser et al. (1978), [An analysis of approximations for maximizing submodular set functions—I — foundation](#). Analyzes greedy approximation for cardinality-constrained submodular maximization. Diminishing returns, monotonicity, and the constraint class are essential when applying greedy selection to datasets or sensors.
- Andreas Krause et al. (2008), [Near-Optimal Sensor Placements in Gaussian Processes: Theory, Efficient Algorithms and Empirical Studies — method](#). Uses mutual-information submodularity for sensor placement, with lazy evaluation and robustness extensions addressing model uncertainty and sensor failures.
- Grigori Călinescu et al. (2011), [Maximizing a Monotone Submodular Function Subject to a Matroid Constraint — theory](#). Combines continuous greedy optimization with rounding to achieve approximation guarantees under matroid constraints. Shows how continuous relaxations can solve discrete selection problems beyond simple cardinality budgets.
- Francis Bach (2013), [Learning with Submodular Functions: A Convex Optimization Perspective — survey](#). Develops connections among submodularity, Lovász extensions, polyhedra, and structured regularization. Distinguishing minimization from maximization prevents importing guarantees into the wrong discrete problem.

VII. Optimization foundations and classical methods

49. Stochastic approximation, SGD, minibatching, and interpolation

- Herbert Robbins and Sutton Monro (1951), [A Stochastic Approximation Method — foundation](#). Introduces recursive root-finding from noisy observations, the foundational template behind stochastic approximation and SGD.
- Boris T. Polyak and Anatoli B. Juditsky (1992), [Acceleration of Stochastic Approximation by Averaging — theory](#). Shows that averaging stochastic-approximation iterates can

achieve asymptotically efficient estimation under suitable regularity, reducing sensitivity to step-size tuning.

- Léon Bottou et al. (2018), [Optimization Methods for Large-Scale Machine Learning — survey](#). Connects statistical learning objectives to stochastic optimization, variance reduction, and curvature methods; a broad orientation before selecting a narrow research question.
- Robert Mansel Gower et al. (2019), [SGD: General Analysis and Improved Rates — theory](#). Analyzes arbitrary sampling through expected smoothness, linking step sizes, residual gradient noise, and minibatch selection.
- Sharan Vaswani et al. (2019), [Fast and Faster Convergence of SGD for Over-Parameterized Models and an Accelerated Perceptron — theory](#). Studies strong and weak growth conditions in interpolation settings, explaining when constant-step SGD and acceleration under stronger assumptions can match deterministic rates.
- Siyuan Ma et al. (2018), [The Power of Interpolation: Understanding the Effectiveness of SGD in Modern Over-parametrized Learning — theory](#). Under convexity, smoothness, interpolation, and curvature assumptions, explains fast minibatch SGD and the transition between linear batch scaling and saturation, with explicit quadratic analysis.

50. Finite-sum optimization and variance reduction

- Rie Johnson and Tong Zhang (2013), [Accelerating Stochastic Gradient Descent using Predictive Variance Reduction — method](#). Introduces SVRG’s snapshot control variate, obtaining fast finite-sum convergence without storing a table of all component gradients.
- Aaron Defazio et al. (2014), [SAGA: A Fast Incremental Gradient Method With Support for Non-Strongly Convex Composite Objectives — method](#). Introduces an unbiased table-based variance-reduced estimator with proximal support and guarantees that adapt to strong convexity.
- Mark Schmidt et al. (2017), [Minimizing finite sums with the stochastic average gradient — method](#). Analyzes SAG’s memory of past gradients, demonstrating how finite-sum structure supports faster convergence than generic stochastic gradient methods.
- Zeyuan Allen-Zhu (2018), [Katyusha: The First Direct Acceleration of Stochastic Gradient Methods — method](#). Combines variance reduction with an additional momentum mechanism to accelerate convex finite-sum optimization, including minibatch variants.
- Cong Fang et al. (2018), [SPIDER: Near-Optimal Non-Convex Optimization via Stochastic Path-Integrated Differential Estimator — method](#). Introduces recursive gradient-difference estimation with periodic refreshes, improving nonconvex stochastic and finite-sum stationarity complexity under smoothness and variance assumptions; extends variance reduction beyond strongly convex objectives.

51. Momentum, acceleration, restart, and dynamical systems

- Boris T. Polyak (1964), [Some methods of speeding up the convergence of iteration methods](#) — *foundation*. Develops multistep iteration methods, including heavy-ball momentum, and motivates acceleration through the behavior of linear and quadratic problems.
- Yurii E. Nesterov (1983), [A method of solving a convex programming problem with convergence rate \$O\(1/k^2\)\$](#) — *foundation*. Introduces accelerated gradient iteration with an optimal-order objective convergence rate for smooth convex optimization.
- Weijie Su et al. (2016), [A Differential Equation for Modeling Nesterov’s Accelerated Gradient Method: Theory and Insights](#) — *theory*. Derives an ODE limit for acceleration, yielding interpretable dynamics and restart ideas while inviting careful comparison with discrete algorithms.
- Laurent Lessard et al. (2016), [Analysis and Design of Optimization Algorithms via Integral Quadratic Constraints](#) — *theory*. Uses control-theoretic stability certificates and small semidefinite programs to analyze convergence and robustness of gradient and momentum algorithms.
- Brendan O’Donoghue and Emmanuel J. Candès (2015), [Adaptive Restart for Accelerated Gradient Schemes](#) — *method*. Proposes simple adaptive momentum resets guided by oscillation; heuristic analysis and experiments motivate studying when restart reliably improves acceleration.
- Chaoyue Liu and Mikhail Belkin (2020), [Accelerating SGD with momentum for over-parameterized learning](#) — *counterexample*. Shows stochastic Nesterov momentum need not accelerate SGD and can diverge; MaSS adds compensation and obtains accelerated guarantees for interpolating quadratic problems under specified moment conditions.

52. Mirror descent, non-Euclidean geometry, and adaptive optimizers

- Amir Beck and Marc Teboulle (2003), [Mirror descent and nonlinear projected subgradient methods for convex optimization](#) — *foundation*. Derives mirror descent as nonlinear projected subgradient iteration and explains dimension-sensitive benefits of entropic geometry on the simplex.
- John Duchi et al. (2011), [Adaptive Subgradient Methods for Online Learning and Stochastic Optimization](#) — *method*. Introduces AdaGrad-style geometry learned from past gradients, with regret guarantees and motivation from sparse, unevenly observed features.
- Diederik P. Kingma and Jimmy Ba (2015), [Adam: A Method for Stochastic Optimization](#) — *method*. Introduces exponential first- and second-moment updates widely used in training; read its original convergence claims alongside later counterexamples and repairs.
- Sashank J. Reddi et al. (2018), [On the Convergence of Adam and Beyond](#) — *counterexample*. Exhibits convex examples where Adam fails to converge and proposes AMSGrad’s long-term second-moment memory to address the identified issue.

- Ilya Loshchilov and Frank Hutter (2019), [Decoupled Weight Decay Regularization — method](#). Separates weight decay from adaptive gradient updates and shows why adding an L2 penalty is not equivalent to decoupled decay for adaptive optimizers.
- Ashia C. Wilson et al. (2017), [The Marginal Value of Adaptive Gradient Methods in Machine Learning — counterexample](#). Constructs overparameterized classification examples where adaptive methods select worse-generalizing solutions than SGD, and reports neural-network experiments; these findings do not establish universal inferiority of adaptive optimization.

53. Second-order, quasi-Newton and structured preconditioning

- Dong C. Liu and Jorge Nocedal (1989), [On the limited memory BFGS method for large scale optimization — foundation](#). Develops limited-memory quasi-Newton optimization using a small history of displacement and gradient-difference vectors instead of a dense inverse Hessian.
- James Martens (2010), [Deep learning via Hessian-free optimization — method](#). Makes curvature-based neural-network training practical through Hessian-free linear solves, damping and curvature choices.
- James Martens and Roger Grosse (2015), [Optimizing Neural Networks with Kronecker-factored Approximate Curvature — method](#). Approximates the Fisher matrix with layer-wise Kronecker factors, trading curvature accuracy for efficient natural-gradient updates.
- Vineet Gupta et al. (2018), [Shampoo: Preconditioned Stochastic Tensor Optimization — method](#). Uses tensor structure to maintain smaller matrix preconditioners and connects stochastic convex guarantees with deep-learning experiments.

54. Nonsmooth, composite, and weakly convex optimization

- Amir Beck and Marc Teboulle (2009), [A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems — method](#). Introduces FISTA, combining inexpensive proximal-gradient updates with accelerated objective convergence for convex composite inverse problems.
- Neal Parikh and Stephen Boyd (2014), [Proximal Algorithms — survey](#). A practical introduction to proximal operators, splitting methods, and common nonsmooth penalties; useful for implementing and interpreting composite algorithms.
- Damek Davis and Dmitriy Drusvyatskiy (2019), [Stochastic Model-Based Minimization of Weakly Convex Functions — theory](#). Unifies stochastic subgradient, proximal-point, and prox-linear updates through descent of a Moreau-envelope stationarity measure for weakly convex objectives.
- Jérôme Bolte et al. (2014), [Proximal alternating linearized minimization for nonconvex and nonsmooth problems — theory](#). Introduces PALM and establishes convergence of bounded iterates to a critical point using the Kurdyka–Łojasiewicz property for broad nonconvex composite models.

- Damek Davis et al. (2018), [Subgradient Methods for Sharp Weakly Convex Functions — theory](#). Proves local linear convergence for selected subgradient step rules when weak convexity is combined with sharp growth and sufficiently close initialization; applies to suitable recovery formulations.

55. Sparse and composite optimization: proximal methods and coordinate descent

- Amir Beck and Marc Teboulle (2009), [A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems — method](#). Introduces FISTA, combining inexpensive proximal-gradient updates with accelerated objective convergence for convex composite inverse problems.
- Neal Parikh and Stephen Boyd (2014), [Proximal Algorithms — survey](#). A practical introduction to proximal operators, splitting methods, and common nonsmooth penalties; useful for implementing and interpreting composite algorithms.
- Paul Tseng (2001), [Convergence of a Block Coordinate Descent Method for Nondifferentiable Minimization — theory](#). Establishes stationarity under separability and regularity conditions for nonsmooth block updates. Offers a precise checklist for diagnosing why coordinate methods can stall outside their assumptions.
- Yurii Nesterov (2012), [Efficiency of Coordinate Descent Methods on Huge-Scale Optimization Problems — theory](#). Analyzes randomized coordinate updates and acceleration for large convex problems. Per-coordinate smoothness and inexpensive partial derivatives are central to translating iteration bounds into useful computational savings.

56. Projection-free and oracle-based constrained optimization

- Martin Jaggi (2013), [Revisiting Frank-Wolfe: Projection-Free Sparse Convex Optimization — method](#). Unifies projection-free sparse optimization through linear minimization oracles, approximate subproblems and computable duality-gap certificates.
- Simon Lacoste-Julien and Martin Jaggi (2015), [On the Global Linear Convergence of Frank-Wolfe Optimization Variants — theory](#). Analyzes global linear convergence of away-step and related variants, making the role of feasible-set geometry explicit.
- Sashank J. Reddi et al. (2016), [Stochastic Frank-Wolfe Methods for Nonconvex Optimization — method](#). Extends projection-free optimization to stochastic and finite-sum nonconvex problems, using variance reduction to improve stationarity guarantees.

57. Nonconvex first-order complexity and escaping saddles

- Saeed Ghadimi and Guanghui Lan (2013), [Stochastic First- and Zeroth-Order Methods for Nonconvex Stochastic Programming — theory](#). Establishes nonasymptotic stationarity guarantees for randomized stochastic gradient methods, giving a foundational analysis for smooth nonconvex stochastic optimization.
- Jason D. Lee et al. (2016), [Gradient Descent Only Converges to Minimizers — theory](#). Under strict-saddle and regularity assumptions, randomly initialized gradient descent almost surely avoids strict saddles; this asymptotic result does not supply a rapid escape bound.

- Chi Jin et al. (2017), [How to Escape Saddle Points Efficiently](#) — *method*. Adds controlled perturbations to gradient descent and proves finite-time approximate second-order stationarity under smoothness and Hessian-Lipschitz assumptions.
- Yair Carmon et al. (2020), [Lower bounds for finding stationary points I](#) — *theory*. Constructs oracle-complexity lower bounds for smooth nonconvex stationarity, clarifying which improvements require assumptions beyond generic smoothness.